

Rahil Parikh

Philadelphia, PA | +1 774-243-4073 | rahilpar@engineering.upenn.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Robotics researcher working on active perception and end-to-end sensorimotor policies — learning not just how to act, but where to look. Currently building a quadruped in NVIDIA Isaac Lab whose pan-tilt depth camera is itself policy-controlled, driven by an information-gain reward grounded in foothold prediction. Backed by nearly three years shipping production real-time C++ control software: a 47,000-line closed-loop stack with FPGA safety interlocks running daily on Jetson hardware at customer research facilities — the systems layer that sim-to-real transfer actually runs on. Dual B.S. Robotics Engineering & Computer Science, WPI (with distinction); incoming M.S.E. Robotics, Penn GRASP, Fall 2026.

EDUCATION

University of Pennsylvania, Philadelphia, PA

Incoming Fall 2026

M.S.E. in Robotics (GRASP Lab), School of Engineering and Applied Science

Expected 2028

- **Planned coursework:** CIS 5800 Machine Perception, MEAM 5170 Control & Optimization with Applications in Robotics, ESE 6500 Learning in Robotics, Physical Intelligence

Worcester Polytechnic Institute, Worcester, MA

Aug 2019 – May 2023

Dual B.S. in Robotics Engineering and Computer Science — With Distinction, GPA: 3.8/4.0

- **Coursework:** Control Engineering, Robot Dynamics, Unified Robotics I–IV, Computer Vision, Machine Learning, Artificial Intelligence (graduate-level), Linear Algebra, Probability, Embedded Systems
- WPI Presidential Scholarship; Dean's List; SWEET Fellowship (research dedication and technical communication); Upsilon Pi Epsilon

RESEARCH & ROBOTICS PROJECTS

GazeGait — Active Perception for Legged Locomotion | *Isaac Lab, PyTorch, RSL-RL* Apr 2026 – Present

- Treating **gaze as an action**: a single recurrent policy on ANYmal-C outputs 12 joint targets *and* pan/tilt rate commands for an onboard depth camera, so the agent learns where to look and how to walk jointly — prior vision-based locomotion mounts the camera passively
- Designed an **information-gain intrinsic reward** grounded in a terrain-predictor head's error on future foothold heights: looking is rewarded only when it improves the policy's own estimate of upcoming contact geometry — task-grounded rather than generic curiosity (ICM/RND)
- Exploit-proofed the intrinsic reward through stop-gradient isolation, a telescoping delta formulation, and per-environment EMA baselines, preventing the predictor from being degraded to farm reward
- Built a five-condition ablation and a scientific evaluation suite measuring foothold anticipation distance, saccade phase-locking against the gait cycle, and gaze-terrain mutual information — testing predictions from human gaze behavior on rough terrain (Matthis et al., *Current Biology* 2018)

Quadruped Locomotion via Deep RL | *Isaac Lab, RSL-RL, PPO, PyTorch, A100*

Nov 2025 – Present

- Trained four PPO locomotion policies on ANYmal-C across 4,096 parallel environments and a flat/slope/stairs terrain curriculum, verifying genuine trotting gaits in all four (duty cycle 0.33–0.62, diagonal foot correlation –0.56 to –0.90) through gait analysis on held-out terrain rather than reward curves alone
- Designed a leave-one-out reward ablation quantifying the **reward-behavior gap**: removing foot slip raised total reward +32.6% while halving walking distance, and terrain clearance proved most valuable at –20.2% task return when removed — direct motivation for moving reward design out of hand-tuning
- Diagnosed a silent checkpoint-loading failure in rsl-rl v5 where mismatched state-dict keys loaded untrained weights, masking near-zero locomotion behind a standing-bonus reward; resolved pipeline-wide via a shared runner utility

HURON — Full-Size Bipedal Humanoid | *Senior Capstone (MQP) | C++, ROS, Gazebo* Aug 2022 – May 2023

- Led a 12-person interdisciplinary team building a self-balancing bipedal humanoid for dynamic walking; secured \$15,000 in prototype funding through a faculty advisory pitch cycle
- Owned the controls stack for an underactuated biped: foot-force stability-margin analysis for disturbance response and inverse-kinematics gait generation with ZMP feedback, achieving sub-50 ms reactive balance on hardware and sustained partial walking
- Built the hardware-software interface end to end — actuator calibration, contact and force sensing, and the real-time loop coupling balance feedback to joint commands

MotionMD — Markerless Human Motion Capture | *Python, MediaPipe, OpenCV*

2024

- Built a monocular video pipeline extracting clinically meaningful gait metrics from human motion via landmark estimation, smoothing, and automated reporting — capturing human movement in the wild with commodity sensing, no marker suit or instrumented lab
- Owned the data pipeline end to end: capture protocol, pre-processing, landmark post-processing, quality validation, and failure-mode evaluation across occlusion, viewpoint, and lighting

PROFESSIONAL EXPERIENCE

Control Systems Software Engineer | *Bridge12 Technologies, Natick, MA*

Sep 2023 – Apr 2026

- Author and owner of a 47,000-line Qt/C++ real-time control system coordinating 8+ heterogeneous instruments through a multi-device finite state machine; deployed in production and operated daily by non-engineers at customer research facilities
- Enforced safety through FPGA hardware interlocks arbitrating against the software control law — hard constraints override nominal control deterministically inside the loop's timing budget, protecting high-power instrumentation from damage
- Delivered hard real-time search-and-track over 100,000+ data points in under 75 ms via spatial hashing and k-d trees, holding deterministic timing under full system load
- Cut closed-loop latency 70% (5.0 s $\rightarrow$ 1.5 s) by replacing synchronous polling with a ZeroMQ asynchronous publisher and Kalman-filtered distributed state estimation, raising usable control bandwidth
- Engineered Yocto-based Linux distributions for NVIDIA Jetson Nano and Orin NX, tuning kernel and boot configuration to eliminate latency jitter on embedded control hardware — the deployment layer a learned policy has to survive
- Led on-site deployment, debugging, and operator training across 5+ research sites (incl. Columbia University, Havemeyer Hall), converting ambiguous physics requirements into shippable closed-loop systems

Software Engineering Intern | *Ekotrope, Boston, MA*

May – Oct 2022

- Raised MariaDB transaction throughput 20% through query and schema redesign on a physics-based building energy modeling backend; reduced modeling defect rate 18% with JUnit suites validated against analytical baselines

ADDITIONAL PROJECTS

Autonomous Mobile Robot Navigation | *C++, ROS, LiDAR, AMCL, GMapping*

2021

- Built a full autonomy stack (SLAM, particle-filter localization, A*/move_base planning) with Kalman-filtered LiDAR-odometry fusion; demonstrated collision-free indoor navigation as lead software developer

Vision-Guided 6-DOF Manipulator | *C++, MATLAB, OpenCV*

2021

- Built a forward/inverse-kinematics pipeline with an OpenCV pose-estimation front end driving a manipulator to autonomous color-sorted pick-and-place

TECHNICAL SKILLS

Robot Learning & Simulation: NVIDIA Isaac Lab, Isaac Gym, MuJoCo, Gazebo, RSL-RL, PPO, massively parallel RL, reward shaping, intrinsic motivation, terrain curricula, sim-to-real transfer, domain randomization, imitation learning
Perception & Vision: PyTorch, OpenCV, MediaPipe, depth and point cloud processing, LiDAR sensor fusion, camera calibration, pose estimation, active/interactive perception, vision-conditioned policy learning

Control & Estimation: real-time and safety-critical control, deterministic loop design, feedback control, PID, Kalman filtering, state estimation, contact dynamics, inverse kinematics, ZMP balance control, MPC formulation, SO(3)/SE(3) Lie groups

Embedded & Systems: C++ (production, multithreaded), NVIDIA Jetson (Nano, Orin NX), embedded Linux, Yocto, FPGA hardware interlocks, Qt/QML, ZeroMQ, ROS/ROS2, Docker, Git

Languages: C++, Python, MATLAB, CUDA, Java, SQL

Currently Studying: Vision-Language-Action models, diffusion policies, world models as learned reward, micro Lie theory for state estimation

SELF-STUDY & SCHOLARSHIP

- **Micro Lie Theory for Robotics** (Solà et al.) — working knowledge of SO(3)/SE(3) Lie groups, exponential/log maps, adjoint, right/left Jacobians, and EKF/SLAM/VIO applications
- **Google DeepMind “Measuring Progress Toward AGI” Kaggle Hackathon** — authored benchmarks in the Metacognition and Executive Functions tracks
- **Reading:** Agarwal et al. (CoRL 2022) egocentric-vision legged locomotion; Loquercio et al. (ICRA 2023) cross-modal supervision for visual locomotion; Pfeiffer et al. (PLOS ONE 2022) visual attention in drone racing; Singh et al. (ICRA 2026) predictive human-motion models as learned reward